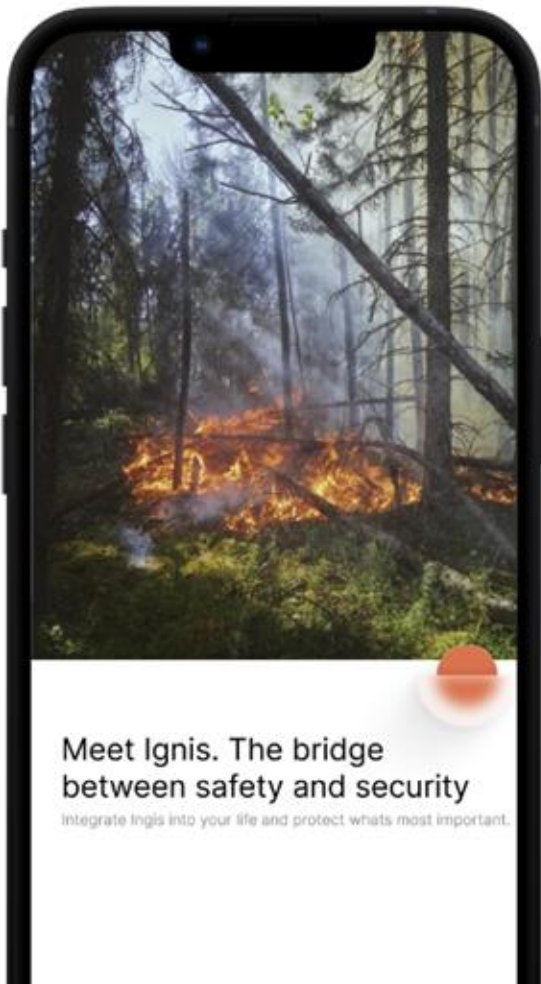
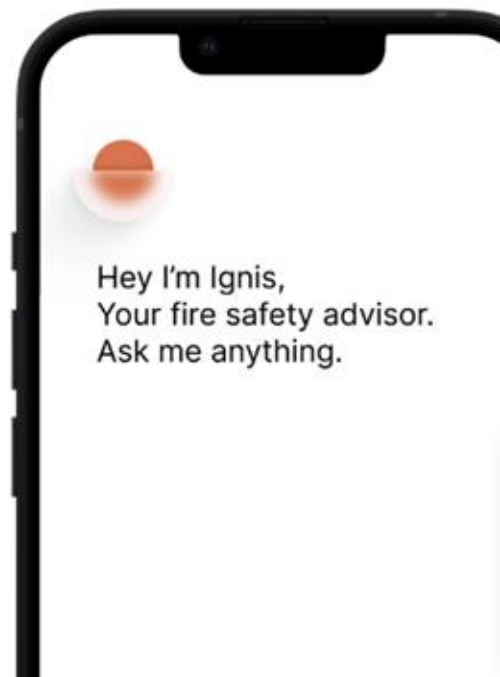




Meet Ignis. The bridge
between safety and security

Integrate Ignis into your life and protect what's most important.



Hey I'm Ignis,
Your fire safety advisor.
Ask me anything.

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Problem

WILDFIRES IN NUMBERS

Wildfires are a major concern, with over 12,972 wildfires burning 351,821 acres in the US alone. As more people move into high-risk areas, it's important to focus on education and preparedness to reduce future risks and save lives.

2400

Deaths between
1998-2017 caused
by wildfires

6,200,000

People affected by
wildfires between
1998-2017

85%

Percent of wildfires
caused by humans in
California and Oregon

Let's Face the Facts

**ONLY 26% OF HOUSEHOLDS IN THE U.S.
HAVE A FIRE SAFETY PLAN**

Source: American Red Cross

MEET

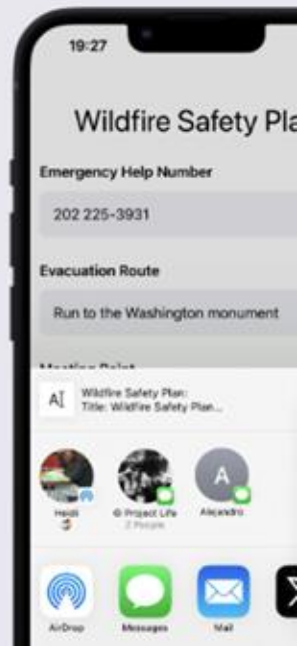
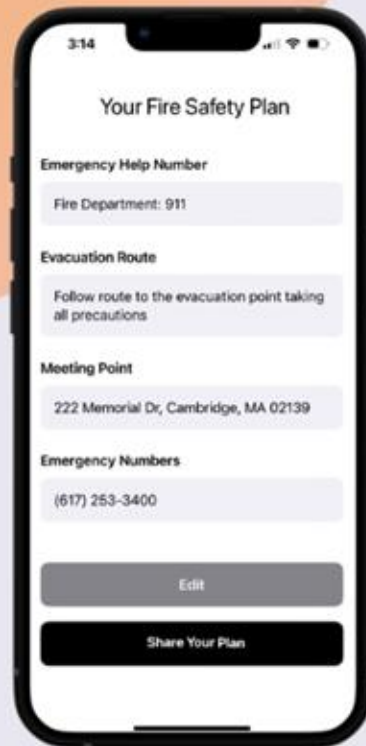
IGNIS

Let's Change That | myCircle

Fire Safety Plan

- **A fire safety plan is crucial**
- With the help of Ignis, families can create and customize their own plans based on their unique needs and circumstances.

Ignis allows families to be better prepared, stay calm, and have a greater chance of staying safe when it most.



Solution V14

AI POWERED HELP

- In the future Ignis will incorporate **Machine Learning** technology to provide personalized guidance and support during fire emergencies, helping users make informed decisions quickly.
- Ignis will also serves as a chatbot to answer any fire related questions that you may have.

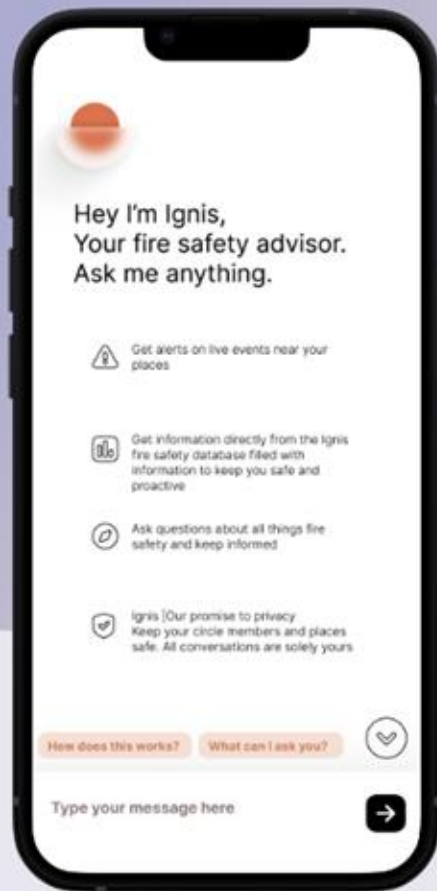


Solution

LOCATION SPECIFIC

At Ignis, we provide customized information on fire risks and safety measures based on the user's **exact location**. Our aim is to equip users with the most **relevant** and **helpful information** possible.

We go beyond generic fire tips.



Solution

**EARLY DETECTION. EARLY
WARNING. IGNIS -
PROTECTING WHAT
MATTERS MOST.**



TRACK + DETECT

Currently **National Evacuation Orders** give 24 hours notice of fire impact.

Ignis allows users to be proactive, informing users within two hours of a fire's detection by NASA Satellites.



Solution

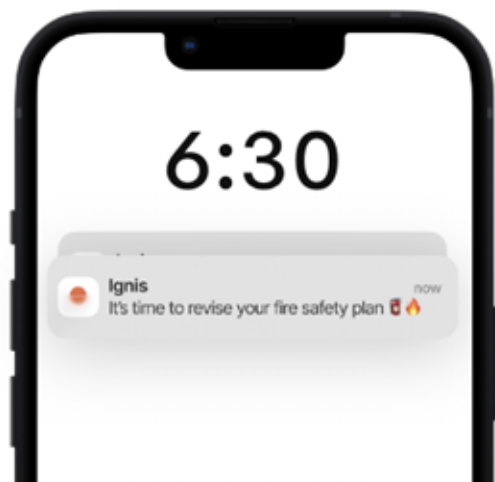
MONITOR MULTIPLE HOMES

Ignis allows users to monitor different homes they are in connection with. This creates a central hub for the safety statuses of both their own, and loved ones homes.



Alerts

IGNIS SENDS REAL-TIME ALERTS TO USERS
ABOUT NEARBY FIRES, EVACUATION ORDERS,
AND THEIR HOME STATUS TO KEEP THEM
INFORMED AND SAFE.





Ignis

now

Red Alert: The detected fire is within 1 mile of **House 1**



Threat level alerts based on the proximity of fire and its potential impact



Ignis

now

It's time to revise your fire safety plan



Quarterly reminders to update your safety plan to reflect your current situation



Ignis

now

A fire has been detected near **House 1** 🚒🔥



Notifications of a **new active fire** to allow you to stay ahead of the flames

Solution

FIRE PREVENTION & EDUCATION





NASA FIRMS API

API

Service	Description
area	Fire detection hotspots based on area, date and sensor in CSV format
countries	List of supported countries
country	Fire detection hotspots based on country, date and sensor in CSV format
data_availability	Date availability of SP and NRT data
kml_fire_footprints	KML fire detection footprints
map_key	Setup MAP_KEY

- NASA FIRMS API (Fire Information for Resource Management System):
 - Data Source: Provides near real-time global fire data, including information on active fires.
 - Satellite Data: Visible Infrared Imaging Radiometer Suite (VIIRS) sensor aboard the joint NASA/NOAA Suomi National Polar-orbiting Partnership (Suomi NPP) and NOAA-20 satellites.
- Relevance
 - Real-Time Fire Information
 - Mapping and Visualization: Supports mapping and visualization features
- Use Cases
 - Displaying Current Fires: Presents users with the latest information on active fires worldwide.
 - Alerts and Notifications: Enables alerts and notifications based on real-time fire data.
 - Historical Analysis: Facilitates historical analysis of fire patterns and trends for a more informed understanding.



NASA FIRMS API

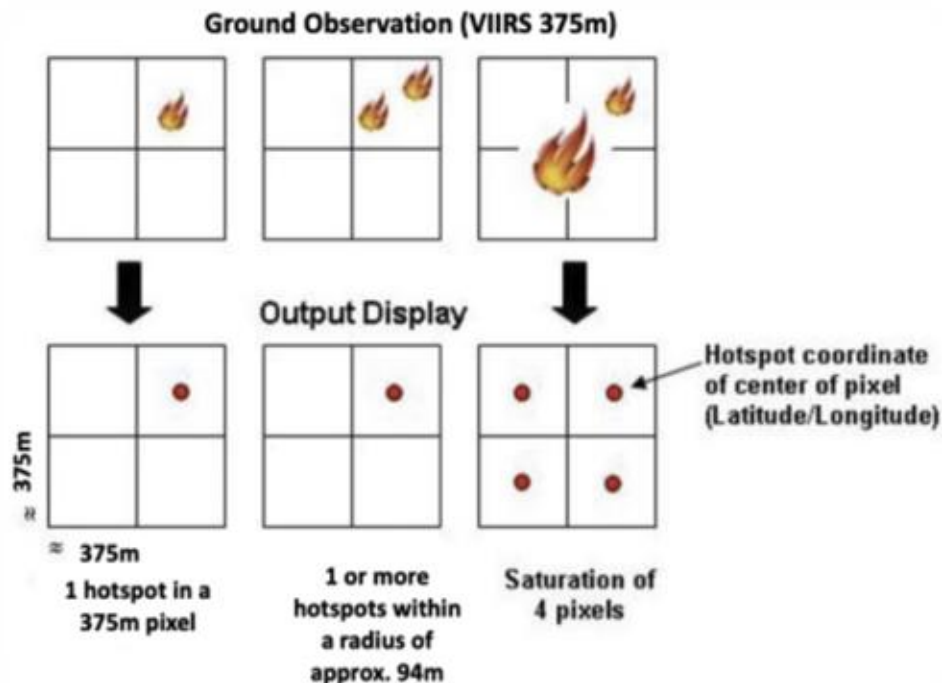
```
url = 'https://firms.modaps.eosdis.nasa.gov/api/area/csv/' + MAP_KEY + '/VIIRS_NOAA20_NRT/-160,23,-66,71/1'
```

API CSV Output:

latitude, longitude, bright_ti4, scan, track, acq_date, acq_time, satellite, instrument, confidence, version, bright_ti5, frp, daynight

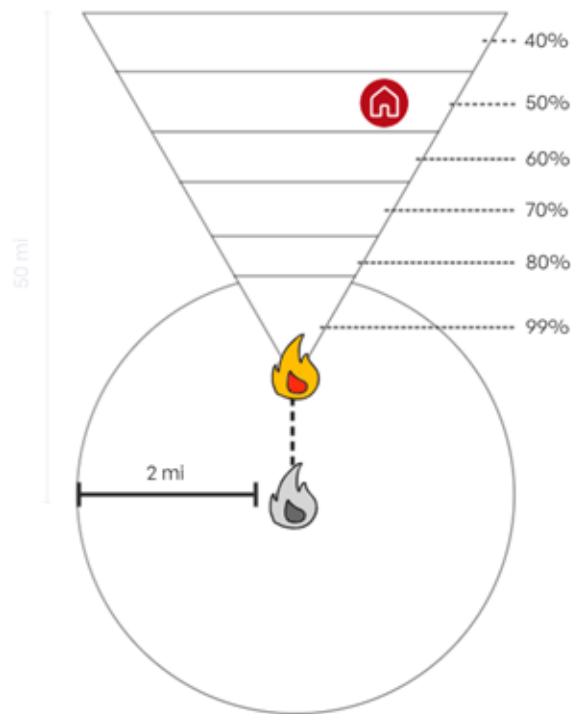
- United States
- Recurring Google Cloud Function
- Converted to JSON
- Stored in Firebase Realtime Database

NASA FIRMS (VIIRS NOAA20)



THE ALGORITHM

- A function analyzes the movement of fires by creating a 2-mile radius around each old fire's location and checking for new fires within this range.
- If found, it calculates the distance between them and projects a vector extending 100 miles past the new fire.
- A cone with a 40-mile radius and 100-mile height is formed, aligned with the vector's angle to encompass potential fire spread.
- The user's address is then checked for inclusion in the cone, and a risk percentage is calculated based on its distance from the cone's tip.



Code Snippets

isWithinCircle()

```
func isWithinCircle(location: CLLocation, center: CLLocation, radius: Double) -> Bool {
    let locationCoordinate = CLLocationCoordinate(location.latitude, longitude: location.longitude)
    let centerCoordinate = CLLocationCoordinate(center.latitude, longitude: center.longitude)

    let distanceMiles = locationCoordinate.distanceFrom(centerCoordinate) / 1609.34 // Convert meters to miles

    return distanceMiles < radius
}
```

isAddressInCone()

```
func isAddressInCone(address: String, vector: (dx: Double, dy: Double), addressLatitude: Double, addressLongitude: Double) -> Bool {
    let coneHalfAngle = 0.343799

    let vectorAngle = atan2(vector.dy, vector.dx)
    // Make sure angle is 0 <= theta < 360
    let vectorAngleNormalized = vectorAngle + 0.5 * vectorAngle + 3.14159 // 180 degrees

    let addressVectorAngle = atan2(addressLatitude - vector.dy, addressLongitude - vector.dx)

    let angleDifference = abs(vectorAngleNormalized - addressVectorAngle)

    return angleDifference < coneHalfAngle
}
```

calculatePercentageChange()

```
func calculatePercentageChange(address: String, vector: (dx: Double, dy: Double), addressLatitude: Double, addressLongitude: Double, return: Double) -> Double {
    // Check if the address is within the cone
    let isWithinCone = isAddressInCone(address, vector, addressLatitude, addressLongitude)

    if isWithinCone {
        let distance = sqrt(pow(addressLatitude - vector.dy, 2) + pow(addressLongitude - vector.dx, 2))

        let percentageChange = (distance / radius) * 100

        print("Address is in cone, percentage change: ", percentageChange)
        return percentageChange
    } else {
        return 0.0
    }
}
```

BUILT WITH

- Swift/SwiftUI - Backend/Frontend Language
- Python - Backend language
- NASA FIRMS API - Wildfire data source
- MapKit - Map Framework
- Firebase Firestore - Cloud database used for storing user info/authentication
- Firebase Realtime - Cloud realtime database used for storing API data
- Google Cloud Functions - Cloud functions used for deploying/scheduling reoccurring API call
- CoreLocation - Geocoding
- Foundation - Base Framework
- UIKit - Frontend Framework
- Requests - HTTP requests Python library
- CSV - CSV tools, standard Python library
- JSON - JSON tools, standard Python library

ROADMAP

Whats next for Ignis



myCircle

Adding a community page to allow users to discuss and seek support during a crisis



Smart Hub Integration

Allowing deeper integration into more homes



Expansion to all major catastrophes

Including tsunamis, volcanoes, hurricanes, cyclones, blizzards



Companion Lamp

To bring Ignis into homes. The Companion lamp will be the start of our Ignis line of Products